Jack Deng

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Work Rights: Temporary Graduate visa (subclass 485) - full work rights; valid until 31 July 2028

PROFESSIONAL SUMMARY

Highly accomplished Software Engineer with 2 years of experience at ByteDance, specializing in architecting and deploying scalable, cloud-native microservices. Proficient in Python, AWS, and full-stack development (ReactJS, .NET concepts). Proven ability to drive end-to-end solutions, optimizing high-throughput systems by 40%+ and ensuring 99.99% uptime in large-scale, multi-cloud environments.

WORK EXPERIENCE

Full-Stack Software Engineer 2022.07 - 2024.06

ByteDance | Beijing, China

- Architected and scaled high-concurrency backend microservices using Python and Node.js, supporting 100M+ DAU and maintaining 99.99% system uptime, directly addressing the need for robust, scalable backend development.
- Designed and implemented Infrastructure as Code (IaC) pipelines using AWS services and Terraform principles, automating deployment across multi-cloud environments and ensuring consistent, repeatable infrastructure provisioning.
- Optimized complex data pipelines and API endpoints by implementing caching strategies and performance tuning, resulting in a quantifiable 45% reduction in average API latency at ByteDance's massive scale.
- Developed full-stack features using ReactJS and modern frameworks, integrating robust unit and integration testing (Jest/xUnit concepts) into automated CI/CD pipelines, which accelerated feature delivery speed by over 40%.

PROJECTS

Enterprise RAG Knowledge Base & Intelligent Support Agent (RAG Knowledge & Support Agent) 2025 – Present

Lead System Architect & Full-Stack Developer | Python, FastAPI, LangChain, PostgreSQL (pgvector), ChromaDB, Celery, Redis, React, TypeScript, Docker

- RAG Retrieval & Chunking: Architected an end-to-end Enterprise RAG platform featuring intelligent document chunking, hybrid semantic search, and vector indexing using pgvector & ChromaDB with sub-second retrieval latency.
- Asynchronous Pipelines & Streaming: Constructed high-throughput asynchronous task queues using Celery and Redis for background document ingestion, paired with FastAPI WebSockets/SSE for real-time markdown streaming responses.
- Production Engineering & Deployment: Designed an intuitive React/TypeScript interface with full Markdown/LaTeX rendering, enforced role-based access control (RBAC), and containerized the multi-service architecture with Docker Compose.

LLM-Based Constraint-Aware Intelligent Trip Planning Mar 2026 – Jun 2026

Lead System Engineer & Researcher | Master's Capstone | Python, LangGraph, Constraint Satisfaction (Z3/OR-Tools), Large Language Models, FastAPI, Redis

- State-Graph Agent Architecture: Engineered a multi-agent orchestration pipeline using LangGraph, decomposing complex travel constraints (budget limits, temporal windows, geospatial proximity) into directed acyclic state graphs.
- Deterministic Constraint Pruning: Integrated deterministic constraint satisfaction algorithms (branch-and-bound) with LLM generative proposals, eliminating hallucinated conflicts and reducing constraint violation rates from 34% to 0%.
- High-Throughput Execution: Optimized graph state serialization and Redis caching to execute end-to-end multi-day trip planning in sub-200ms with 100% mathematical compliance.

EDUCATION

The University of Queensland Jul 2024 – Jul 2026

Master of Computer Science (Management)

GPA: 5.5/7.0 | Coursework: Advanced Database Systems, Cloud Computing, High-Performance Systems, Machine Learning, Software Architecture

TECHNICAL SKILLS

Core Languages: Python, C#, TypeScript, JavaScript, Node.js, Go, C++, SQL

Frontend & Web: React.js, Next.js, Vue.js, Tailwind CSS, Redux/Zustand, Responsive UI

Backend & Microservices: ASP.NET Core (.NET 8+), Python, Microservices Architecture, Entity Framework Core, FastAPI, RESTful APIs

Database & Storage: PostgreSQL, MySQL, Redis, MongoDB, ElasticSearch, pgvector

Cloud & DevOps: AWS (ECS, S3, RDS), CI/CD (GitHub Actions), Docker, Kubernetes, Linux